

Internal Assessment Question Paper – 1

Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of CSE

Programme: B. E**Course : Discrete Mathematical Structures****Course Code : CS35****Section: A, B, C****Max Marks: 30****Term: Oct 22 to Jan 2023****CIE : 1****Sem: 3****Time: 9.30 to 10.30****Date: 21-12-2022****Portions for Test: L1-L12****Instructions to Candidates: Question 1 is compulsory**

1. Answer any one full question from 2 or 3. Each Question carries 15M.
2. Mobiles, smart watches or any electronic gadgets are strictly banned.

Sl#	Question	Marks	Bloom's Level	CO Mapping
1	a) i. Differentiate between a statement and open statement with an example ii. Describe whether the following argument is logically correct or not? i. Socrate says: "If I'm guilty, I must be punished; I'm guilty. Thus I must be punished." ii. "If I'm guilty, I must be punished; I'm not guilty. Thus I must not be punished."	5	Understand	CO1
	b) Define a partial order on a set A with an example. Construct the Hasse diagram for the divides relation on following set $A = \{2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72\}$.	5	Apply	CO2
	c) Determine whether the following argument is valid or not with suitable reasons $\forall x [p(x) \vee q(x)]$ $\exists x \sim p(x)$ $\forall x [\sim q(x) \vee r(x)]$ <u>$\forall x [s(x) \rightarrow \sim r(x)]$</u> $\therefore \exists x \sim s(x)$	5	Apply	CO1
2	a) i. Given the following propositional statement, construct a truth table and state whether it is a tautology or not. $(\neg p \wedge \neg(p \rightarrow q))$ ii. Find the negation of the proposition "If the sun is shining, then I am going to the ball game."	4	Understand	CO1
	b) i. Simplify the following network using laws of logic	6	Apply	CO1

	ii. Verify argument is valid or not <i>“If I get my Christmas bonus AND my friends are free, I will take a road trip with my friends. If my friends don't find a job after Christmas, then they will be free. I got my Christmas bonus and my friends did NOT find a job after Christmas. Therefore, I will take a road trip with my friends!”</i>			
	c) Define an equivalence relation with an example. Let $A = \{1, 2, 3, 4, 5\}$, give an example of a mapping which is (i) neither symmetric nor antisymmetric (ii) anti-symmetric and reflexive but not transitive (iii) Transitive and reflexive but not anti-symmetric. (iv) reflexive, symmetric but not transitive (v) reflexive, transitive but not symmetric	5	Understand	CO2
3	a) Recall “Rule of Disjunctive syllogism” and “Domination law” and show that they are Tautologies	4	Remember	CO1
	b) i. Prove the following logical equivalence without using truth table $[(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r)] \Leftrightarrow p \wedge q$ ii. Establish the validity of the following argument by using contradiction. $\begin{array}{l} \sim p \leftrightarrow q \\ q \rightarrow r \\ \hline \sim r \\ \hline \therefore p \end{array}$	7	Apply	CO1
	c) Let $A = \{2, 3, 4, 6, 8, 12, 16\}$. aRb if and only if ‘a divides b’ (i) Write R (ii) Give matrix of R (iii) Draw digraph of R (iv) Indicate the properties of the relations it holds?	4	Apply	CO2

Course Outcomes meant to be assessed by the IA Test:

- CO1: Understand the notion of mathematical logic & proofs and be able to apply them in problem solving.
- CO2: Solve problems which involve discrete data structures such as relations and discrete functions